Fiscal Year:	FY 2026	**Task Last Updated:**	FY 08/30/2025
PI Name:	Yang, Jiseon Ph.D.		
Project Title:	Microbial Social Behavior and Heritable Genetic or Epigenetic Changes Affected by the Spaceflight Environment: Understanding the Evolution of Microbial Interactions During Spaceflight		
Division Name:	Space Biology		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:			
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	(1) Cell & Molecular Biology (2) Microbiology		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	(1) Translational (Countermeasure) Potential		
PI Email:	jyang41@asu.edu	**Fax:**	FY
PI Organization Type:	UNIVERSITY	**Phone:**	415-308-2322
Organization Name:	Arizona State University		
PI Address 1:	1001 S McAllister Avenue		
PI Address 2:	The Biodesign Institute		
PI Web Page:			
City:	Tempe	**State:**	AZ
Zip Code:	85287-5401	**Congressional District:**	9
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2018 Space Biology (ROSBio) NNH18ZTT001N-FG. App B: Flight and Ground Space Biology Research
Start Date:	11/01/2019	**End Date:**	10/31/2026
No. of Post Docs:	0	**No. of PhD Degrees:**	0
No. of PhD Candidates:	0	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	1	**Monitoring Center:**	NASA ARC
Contact Monitor:	Jones, Harry	**Contact Phone:**	650.604.5518
Contact Email:	harry.jones@nasa.gov		
Flight Program:			
Flight Assignment:	NOTE: End date changed to 10/31/2026 per F. Hernandez/ARC (Ed., 3/10/26). NOTE: End date changed to 10/31/2025 per NSSC information (Ed., 9/24/24) NOTE: End date changed to 10/31/2024 per NSSC information (Ed., 9/15/23) NOTE: End date changed to 10/31/2023 per F. Hernandez/ARC (Ed., 2/27/23)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):			
Grant/Contract No.:	80NSSC19K1597		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	September 2023 Update: The Principal Investigator (PI) received official approval from Space Biology on May 30, 2023, to transition the Micro-17 experiment into a ground study (Ed., 9/15/23). Microbes naturally exist as mixed-species communities where they actively communicate and interact with neighboring cells. They adapt and evolve over time within the mixed-species populations. While many studies have shown that the spaceflight environment (and ground-based spaceflight analogs) impacts microbial biological processes, most of the previous experiments were based on single-species cultures. It is still unclear how the spaceflight environment influences microbial social behaviors and how their interaction in the mixed-species community contributes to microbial evolutionary processes in the spaceflight environment. Understanding the effects of the spaceflight environment on the changes in microbial social behavior and the associating evolutionary mechanisms will be necessary for spaceflight design, operations, and astronaut health. It is crucial in regard to preparation for long-duration life support systems during future exploration missions far from Earth. Microbes within the ISS are continuously exposed and adapted to the unique spaceflight environment, which includes microgravity, low fluid shear, and enhanced radiation. A variety of microbial isolates have been recovered from the ISS, including the potable water system. The ISS Water Recovery System recycles urine, condensation, and wastewater. Astronauts rely on the recycled potable water for survival. Most of the microbes recovered from ISS portable water are known biofilm formers, which can increase risks for crew infections or damage to life support system function and integrity. Therefore, it is critical to understand how microbes in the ISS water system adapt and evolve over time. The purpose of the study is to understand the microbial evolution associated with interspecies interaction and to examine the influence of the spaceflight environment on microbial interactions and evolutions in the mixed-species population, by cohort profiling. The hypothesis is that changes in microbial interaction between species of ISS water isolates are associated with microbial genomic evolution during spaceflight, and spaceflight is a contributing factor for the evolution of microbial interaction between different species.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	This study is searching for an understanding by which microorganisms interact and evolve in the context of the complex microbial ecosystem. While immediately applicable to the unique spaceflight environment, it will also be applicable to closed environments in which water can be recycled and reused in terrestrial environments. Overall, this contributes to the creation of healthier and sustainable human built and living environments both in space and on Earth. The study uses the International Space Station as a microbial observatory to study the complex microbial ecosystems and dynamics in human built and living environments, including spaceflights. The study aligns with the research goals in the Space Biology Microbiology Element to determine the effect of spaceflight on life, process, and community dynamics. Completion of this project would 1) provide genomic evidence on the potential microevolution of ISS potable water, 2) provide insight into how microorganisms in the heterogeneous population evolve over time, and 3) determine whether the spaceflight environment is a key contributing factor that influences changes in microbial interaction. The knowledge gained from this study may a) fill knowledge gaps in microbial interspecies dynamics and evolution within the mixed-species population; b) provide novel insight into how the microbial ecological success within the mixed-species population contributes to microbial evolution in human Built Environment, whether in spaceflight or on Earth; and c) provide useful knowledge to support long-duration life support system, particularly, a water recovery system in spaceflight.
Task Progress:	This FY2025 project report covers September 2024 to August 2025. We have made significant progress on each of the proposed aims, with all objectives on track for successful completion. The project is advancing according to plan, and the results to date demonstrate both strong technical execution and scientific impact. • Aim 1: Generated high-quality genomic data and annotations for International Space Station (ISS) Sphingomonas isolates, and initiated genome-to-phenotype correlations using pangenome analysis. Completed extensive interspecies interaction assays between ISS Sphingomonas and Ralstonia insidiosa, generating a robust dataset to evaluate how long-term ISS exposure may influence cooperative or competitive phenotypes. • Aim 2: Completed long-term mixed-species culture experiments profiling population dynamics under spaceflight-analog conditions. Performed statistical analyses to characterize population shifts over time, identifying trends relevant to adaptive processes. Established a comprehensive variant collection from these mixed-species cultures to enable downstream genomic and phenotypic comparisons. Initiated characterization of these variants, including genotypic and phenotypic profiling, and began interspecies interaction assays utilizing the variant strains to assess potential shifts in cooperative or competitive behaviors under spaceflight-analog conditions. • Aim 3: Conducted long-term mixed-species culture experiments with terrestrial (“ground”) isolates to profile and analyze population dynamics in parallel with Aim 2. Completed statistical analysis of population changes and assembled a variant library to enable future comparative analyses of ISS-adapted versus ground-based isolates under spaceflight-analog conditions. Moving forward, we will conduct further analyses to accomplish our proposed aims and achieve our overall goals. Specifically, we plan to advance genomic, transcriptomic, and phenotypic investigations to address the objectives outlined in all three aims. We will also prepare and submit manuscripts to disseminate our findings.
Bibliography Type:	Description: (Last Updated: 09/02/2025)
Articles in Peer-reviewed Journals	Yang J, Kim HD, Barrila J, Lee SH, Nickerson CA, Ott CM, Israel SA, Choukér A, Yang JY. "Navigating mental health in space: gut-brain axis and microbiome dynamics." Exp Mol Med. 2025 Jun;57(6):1152-63. https://doi.org/10.1038/s12276-025-01488-z ; PMID: 40588528; PMCID: PMC12226736 , Jun-2025